

United International University
CSE 4833 Digital Image Processing
Class Test 1 Time: 40 Minutes Marks: 20 SET-C1

1. A home security company is developing a low-power, edge-AI acoustic sensor designed to detect both **glass breakage events** (which generate rapid, high-frequency acoustic transients above **15 kHz**) and **slow, subtle structural flexes** (low-amplitude vibrations caused by someone quietly picking or prying a door frame).

Due to strictly limited onboard SRAM and wireless transmission bandwidth, the hardware team must choose between two microcontroller configurations:

- a. **Sensor System X:** Uses a high sampling rate (**48 kHz**) but a coarse **4-bit** Analog-to-Digital Converter (ADC) for sound amplitude.
 - b. **Sensor System Y:** Uses a lower sampling rate (**8 kHz**) but a fine **16-bit** Analog-to-Digital Converter (ADC) for sound amplitude.
- i. Which system is more likely to suffer from severe spectral aliasing and miss detecting the sharp sound of shattering glass? Provide reasoning for your answer. **[3]**
 - ii. Which system is more likely to fail to detect a faint door-prying attempt due to excessive quantization noise (amplitude distortion)? Provide reasoning for your answer. **[3]**
 - iii. If bandwidth constraint allows upgrading only one parameter (sampling rate or quantization bit depth), which would you prioritize specifically for the glass breakage detection module, and why? **[2]**
2. For your DIP project, you have a dataset of **handwritten English alphabets (capital letters + special characters)**. There are approximately **6** special characters. Each image size is **64x64** pixels, where **6-pixel blocks**(only horizontal) are used to extract features. Each block extracts **2** features. The final output should be one of the alphabets.
How many neurons should we use in the input and output layers? What is the most suitable activation function for the output layer? Show calculations for the number of neurons in the input layer and the output layer(if any). **[4]**

3. Can we use the ReLU activation function in the output layer? Justify your answer with a suitable example. [2]

4. Consider the following **5x5** grayscale image where **V = [40,150]** [6]

220	98	145	110	112
56	165	87	230	148
65	155	96	65	242
196	121	105	210	144
75	85	188	75	98

- (a) Find out and show the connectivity of all the 4-adjacent pixels.
- (b) Find out and show the connectivity of all the 8-adjacent pixels.
- (c) Find out and show the connectivity of all the m-adjacent pixels.